The Supermarket Fixed Point Has No Power Tail: A Disproof of Conjecture 00000005512

GitHub: gaochengzhecpu

Abstract

At load 1/2, the two-choice supermarket fixed point has survival probabilities $q_k = 2^{-(2^k-1)}$. We verify the entire stationary mean-field system, construct the associated normalized probability distribution, and prove that $k^m q_k \to 0$ for every integer m. Thus its queue-length tail has no finite power-law exponent. We also construct the genuine fixed points for every selection count $d \geq 2$: at queue length two their tails are $q_2^{(d)} = 2^{-(d+1)}$, which decay faster than every inverse power of d. Lean verifies both conclusions, the stationary equations, and probability normalization.

Reading and scope. The source describes a hyperbolic-type fixed-point distribution, a power-law tail in the selection count, doubling of the queue-length tail exponent, and a power-tail exponent growing linearly with the number sampled. We disprove genuine finite power-tail assertions in *both* variables: queue length at fixed $d = 2$, and selection count at fixed queue length $k = 2$. A power law here means positive asymptotic constant or positive two-sided comparison with x^{-a} for finite real a. We exclude even the weaker necessary condition of a positive eventual polynomial lower envelope.

There is a different possible reading of the source's wording: the *recursion* for tails takes a d-th power. That statement is correct, and is used below. It describes double-exponential decay in queue length for $d \geq 2$, not a hyperbolic or polynomial tail. No refutation of that corrected recursion-only reading is claimed.

The actual model and its fixed-point equations. Use unit-rate exponential service and arrival load $\rho \in (0, 1)$. Each arrival samples d servers uniformly and joins a shortest queue. Let $q_k(t)$ denote the limiting fraction of queues of length at least k, so $q_0(t) = 1$. The classical mean-field equations are

$$\dot{q}_{k+1} = \rho\big(q_k^d - q_{k+1}^d\big) - (q_{k+1} - q_{k+2}), \qquad k \geq 0. \tag{1}$$

The arrival term is the probability that all sampled queues have length at least k, less the probability that all have length at least $k + 1$. The service term counts queues of length exactly $k + 1$. These are the standard infinite-dimensional equations, not an alternative finite approximation. See Budhiraja, Mukherjee and Wu, *Supermarket Model on Graphs*, Section 2.1, especially equation (2.6), and the introduction's classical clique discussion: arXiv:1712.07607.

We study fixed points of (1), with $q_0 = 1$, $0 \leq q_{k+1} \leq q_k$, and $q_k \to 0$. The term "fixed point" here refers to this standard mean-field equilibrium. A finite-server stationary law or a new proof of the stochastic mean-field limit is not part of this submission.

Theorem 1. *At $\rho = 1/2$ and $d = 2$, the unique fixed point in this tail class is*

$$q_k = 2^{-(2^k-1)}, \qquad k \geq 0.$$

It is the survival tail of a probability distribution on $\mathbb{N}$. For every integer $m \geq 0$, $k^m q_k \to 0$. Consequently it has no finite polynomial tail exponent.

Proof. Define $q_0 = 1$ and $q_{k+1} = \frac{1}{2}q_k^2$. Induction gives the displayed formula, strict positivity, $q_k \le 1$, monotonicity, and $q_k \le 2^{-k}$. The recursion immediately gives

$$\tfrac{1}{2}(q_k^2 - q_{k+1}^2) - (q_{k+1} - q_{k+2}) = 0$$

for every k, so the entire stationary system holds.

For uniqueness, take any fixed point and put $J_k = \frac{1}{2}q_k^2 - q_{k+1}$. Equation (1) says $J_k = J_{k+1}$. Since $q_k \to 0$, also $J_k \to 0$, so every $J_k = 0$. The recursion and $q_0 = 1$ now determine every coordinate.

Set $p_k = q_k - q_{k+1} \ge 0$. Telescoping gives $\sum_{k=0}^{N-1} p_k = 1 - q_N \to 1$, and, for every n, $\sum_{k=n}^{\infty} p_k = q_n$. Thus these are actual survival probabilities of a normalized distribution. Finally,

$$0 \le k^m q_k \le k^m 2^{-k} \longrightarrow 0$$

for every m. If a genuine finite power law $q_k \asymp k^{-a}$ held, choose an integer $m \ge \max(a, 0)$. For all sufficiently large $k \ge 1$, $k^m q_k$ would be bounded below by a positive constant, contradicting the limit just proved. □

Theorem 2. *For every integer $d \ge 2$, at load $1/2$ the sequence defined by*

$$q_0^{(d)} = 1, \qquad q_{k+1}^{(d)} = \tfrac{1}{2}\big(q_k^{(d)}\big)^d$$

is a fixed point of (1) *and the survival tail of a normalized probability distribution. At queue length two,*

$$q_2^{(d)} = 2^{-(d+1)}, \qquad d^m q_2^{(d)} \longrightarrow 0 \quad (d \to \infty)$$

for every integer $m \ge 0$. Hence this dependence on selection count also has no finite power-law tail.

Proof. For $x \in [0, 1]$ and $d \ge 1$, $x^d \le x$. Thus induction gives $0 < q_k^{(d)} \le 1$, $q_{k+1}^{(d)} \le q_k^{(d)}$, and $q_k^{(d)} \le 2^{-k}$, so the tail tends to zero. Substituting the recursion proves every stationary equation. Differences of adjacent tails again form nonnegative masses; telescoping proves total mass one and the survival-tail identity at every level. Two applications of the recursion give $q_1^{(d)} = 1/2$ and $q_2^{(d)} = 2^{-(d+1)}$. Multiplication by any power of d still tends to zero, excluding any positive polynomial lower envelope. □

Formal verification and scope. Lean defines the actual coordinate drift in (1), and its fixed-point predicates include every coordinate equation and the tail boundary conditions. All sequences are infinite functions $\mathbb{N} \to \mathbb{R}$. Mathlib's `HasSum` verifies the normalization and every survival-tail identity. No finite table, numerical iteration, or asymptotic assumption replaces these proofs. The principal results are

`tail_fixed_point`, `fixed_point_unique`,
`tail_closed_form`, `tail_is_survival`,
`weighted_tail_tendsto_zero`, `not_power_tail_conjecture`,
`tailD_fixed_point`, `no_selection_count_power_tail`.

`PolynomialTail` expresses existence of an integer m, a positive c, and an N, such that $n^m q_n \ge c$ for every $n \ge N$. The first negated conjecture is universally quantified over loads and normalized two-choice fixed points. The second negation concerns the genuine family in Theorem 2. Both exclude a necessary condition of any finite real power tail; the elementary real-to-integer exponent comparison in Theorem 1 is a paper explanation of that implication. The stochastic derivation of the mean-field equations and the finite-server limit theorem are cited model background and are not claimed as formalized. Other source clauses need not be settled once the claimed power-tail clause is false.

Source and reproduction. The preserved English and Chinese conjecture appears in `SOURCE.md`; the provenance file records its commit and hash. The project pins Lean 4.19.0 and Mathlib revision

`c44e0c8ee63ca166450922a373c7409c5d26b00b`.

From the `lean` directory run `lake build`, then

`lake env lean -DwarningAsError=true Main.lean`.

All principal axiom reports use only `propext`, `Classical.choice` and `Quot.sound`. Development and self-review were performed by a delegated AI agent; the coordinating agent reviews the result separately. No external independent review is claimed.